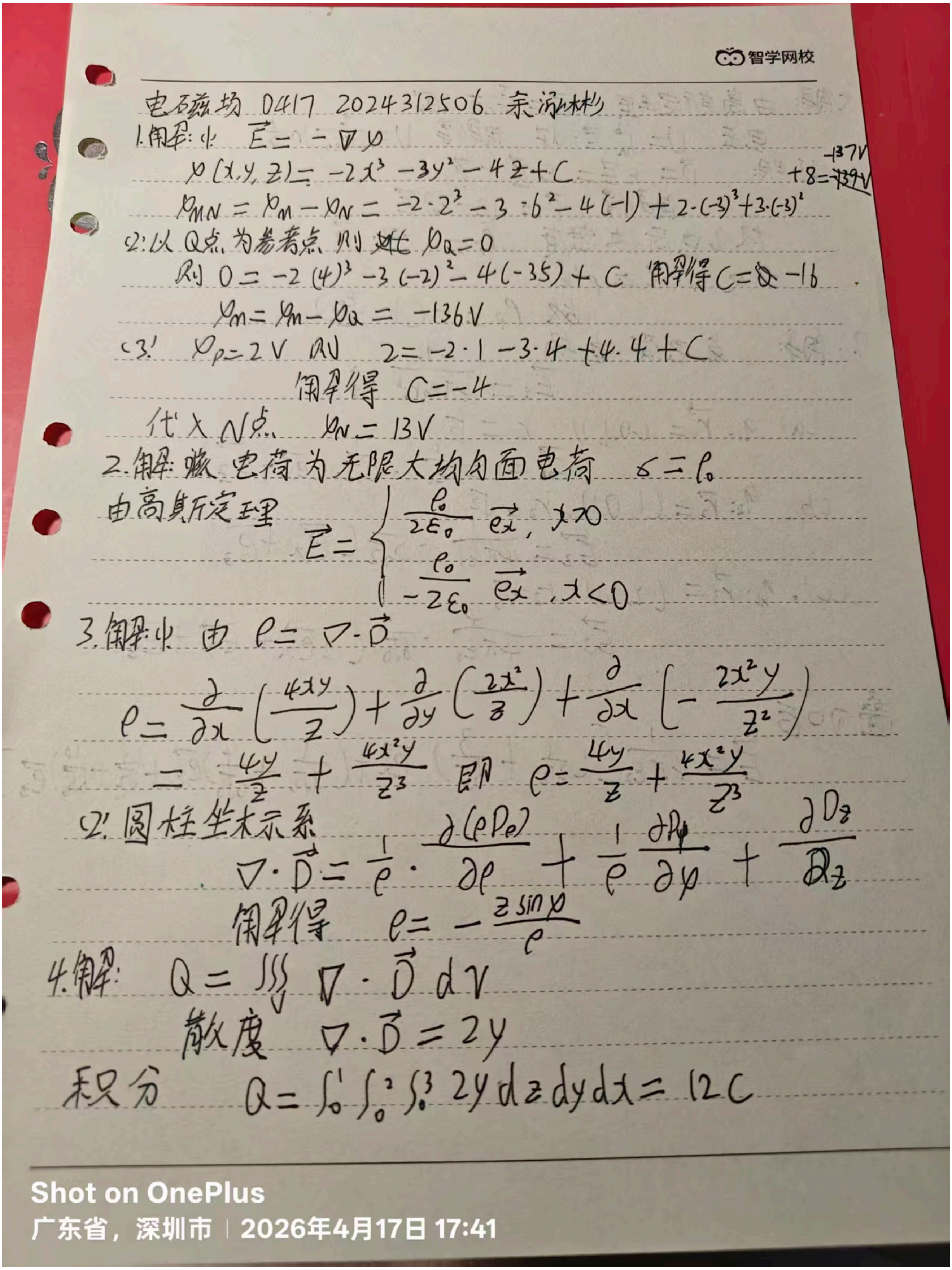

智学网校

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1.解:(1) $\vec{E} = -\nabla\varphi$

$\varphi(x,y,z) = -2x^3 - 3y^2 - 4z + C$ $\qquad$ $-137V$

$+8=$ ~~$-139V$~~

$\varphi_{MN} = \varphi_M - \varphi_N = -2\cdot2^3 - 3\cdot6^2 - 4(-1) + 2\cdot(-3)^3 + 3\cdot(-3)^2$

(2) 以Q点为参考点 则 ~~对t~~ $\varphi_Q = 0$

则 $0 = -2(4)^3 - 3(-2)^2 - 4(-35) + C$ 解得 $C=$ ~~8~~ -16

$\varphi_M = \varphi_M - \varphi_Q = -136V$

(3) $\varphi_P = 2V$ 则 $2 = -2\cdot1 - 3\cdot4 + 4\cdot4 + C$

解得 $C = -4$

代入N点 $\varphi_N = 13V$

2.解: [illegible] 电荷为无限大均匀面电荷 $\sigma = \rho_0$

由高斯定理

$$\vec{E} = \begin{cases} \dfrac{\rho_0}{2\varepsilon_0}\vec{e_x}, & x>0 \\ \dfrac{\rho_0}{-2\varepsilon_0}\vec{e_x}, & x<0 \end{cases}$$

3.解:(1) 由 $\rho = \nabla\cdot\vec{D}$

$$\rho = \frac{\partial}{\partial x}\left(\frac{4xy}{z}\right) + \frac{\partial}{\partial y}\left(\frac{2x^2}{z}\right) + \frac{\partial}{\partial x}\left(-\frac{2x^2y}{z^2}\right)$$

$$= \frac{4y}{z} + \frac{4x^2y}{z^3} \quad 即 \quad \rho = \frac{4y}{z} + \frac{4x^2y}{z^3}$$

(2) 圆柱坐标系

$$\nabla\cdot\vec{D} = \frac{1}{\rho}\cdot\frac{\partial(\rho D_\rho)}{\partial\rho} + \frac{1}{\rho}\frac{\partial D_\varphi}{\partial\varphi} + \frac{\partial D_z}{\partial z}$$

解得 $\rho = -\dfrac{z\sin\varphi}{\rho}$

4.解: $Q = \iiint_V \nabla\cdot\vec{D}\,dV$

散度 $\nabla\cdot\vec{D} = 2y$

积分 $Q = \int_0^1\int_0^2\int_0^3 2y\,dz\,dy\,dx = 12C$

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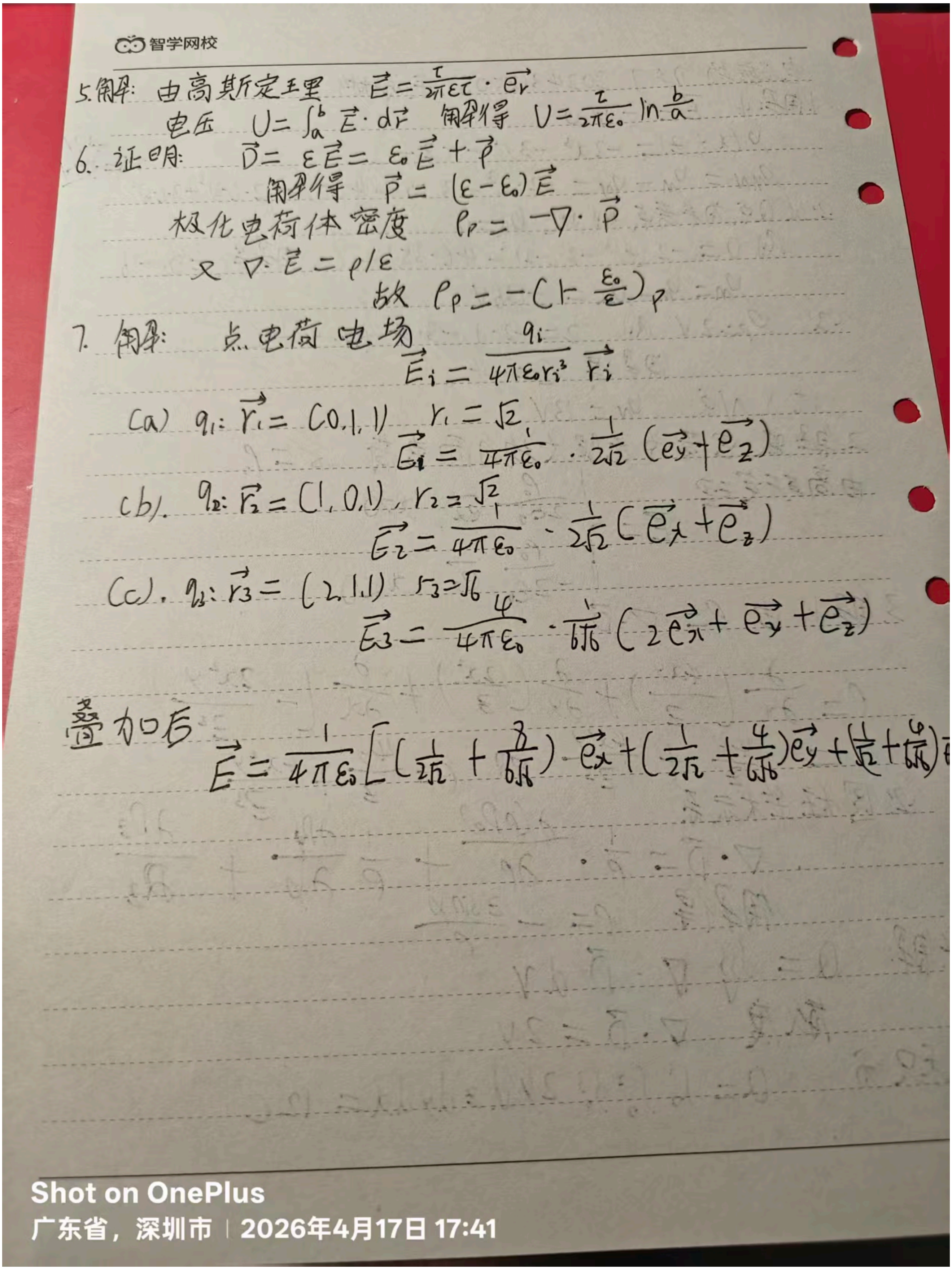

5.解：由高斯定理 $\vec{E}=\frac{\tau}{2\pi\varepsilon\tau}\cdot\vec{e_r}$

电压 $U=\int_a^b\vec{E}\cdot d\vec{r}$ 解得 $U=\frac{\tau}{2\pi\varepsilon_0}\ln\frac{b}{a}$

6. 证明：$\vec{D}=\varepsilon\vec{E}=\varepsilon_0\vec{E}+\vec{P}$

解得 $\vec{P}=(\varepsilon-\varepsilon_0)\vec{E}$

极化电荷体密度 $\rho_P=-\nabla\cdot\vec{P}$

又 $\nabla\cdot\vec{E}=\rho/\varepsilon$

故 $\rho_P=-(1-\frac{\varepsilon_0}{\varepsilon})\rho$

7. 解：点电荷电场

$$\vec{E_i}=\frac{q_i}{4\pi\varepsilon_0 r_i^3}\vec{r_i}$$

(a) q_1: $\vec{r_1}=(0,1,1)$ $r_1=\sqrt{2}$

$$\vec{E_1}=\frac{1}{4\pi\varepsilon_0}\cdot\frac{1}{2\sqrt{2}}(\vec{e_y}+\vec{e_z})$$

(b). q_2: $\vec{r_2}=(1,0,1)$, $r_2=\sqrt{2}$

$$\vec{E_2}=\frac{1}{4\pi\varepsilon_0}\cdot\frac{1}{2\sqrt{2}}(\vec{e_x}+\vec{e_z})$$

(c). q_3: $\vec{r_3}=(2,1,1)$ $r_3=\sqrt{6}$

$$\vec{E_3}=\frac{4}{4\pi\varepsilon_0}\cdot\frac{1}{6\sqrt{6}}(2\vec{e_x}+\vec{e_y}+\vec{e_z})$$

叠加后

$$\vec{E}=\frac{1}{4\pi\varepsilon_0}\left[\left(\frac{1}{2\sqrt{2}}+\frac{8}{6\sqrt{6}}\right)\vec{e_x}+\left(\frac{1}{2\sqrt{2}}+\frac{4}{6\sqrt{6}}\right)\vec{e_y}+\left(\frac{1}{2\sqrt{2}}+\frac{4}{6\sqrt{6}}\right)\right.$$

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